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Report

Exceptionally Preserved 450-Million-Year-Old Ordovician Ostracods with Brood Care

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Summary

Ostracod crustaceans are the most abundant fossil arthropods and are characterized by a long stratigraphic range. However, their soft parts are very rarely preserved, and the presence of ostracods in rocks older than the Silurian period [1–5] was hitherto based on the occurrence of their supposed shells. Pyritized ostracods that preserve limbs and in situ embryos, including an egg within an ovary and possible hatched individuals, are here described from rocks of the Upper Ordovician Katian Stage Lorraine Group of New York State, including examples from the famous Beecher's Trilobite Bed [6, 7]. This discovery extends our knowledge of the paleobiology of ostracods by some 25 million years and provides the first unequivocal demonstration of ostracods in the Ordovician period, including the oldest known myodocope, *Luprisca incuba* gen. et sp. nov. It also provides conclusive evidence of a developmental brood-care strategy conserved within Ostracoda for at least 450 million years.

Results

Ostracods are ubiquitous today in a wide range of freshwater and marine environments and depths. Presumed fossil examples are used extensively in biostratigraphic and paleoenvironmental studies. Most of the tens of thousands of extant ostracod species are benthic or nektobenthic [8]. There is also a minor but important pelagic fauna of ostracods that originated during the Silurian period (~420–443 mega-annum [Ma]) based on both morphological and geological evidence [8–13]. Molecular phylogenetics combined with evidence from fossils calibrates a maximum divergence time for the monophyletic Ostracoda at about 500 Ma during the Cambrian period [14]. Although fossil shells attributed to ostracods occur in countless numbers from the Ordovician period (~443–485 Ma) onward, shells alone are not definitive evidence of affinity [15]: a number of other arthropod taxa have bivalved carapaces. However, fossil ostracods preserving soft parts are extremely rare, especially in the Paleozoic [1–5]. In situ embryos, including possible cases of brooding, are equally scarce in fossil ostracods and other invertebrates [2]. Here, we describe pyritized ostracods preserving soft parts and embryos from the Ordovician Lorraine Group—a stratigraphic unit that has yielded other exceptionally preserved invertebrates [6, 7], most notably the remarkably preserved trilobite *Triarthrus* from Beecher's Trilobite Bed. The significance of this discovery is that it extends our knowledge of the paleobiology of the most abundant group of fossil arthropods by some 25 million years and confirms, for the first time, the presence of ostracods in the Ordovician period. It is the only known fossil invertebrate species that preserves evidence of ovarian-to-juvenile brood care. This discovery confirms that this reproductive strategy, which is found in myodocope ostracods today, was present in the earliest-known representatives of the group 450 million years ago.

The ostracods are held by the Yale Peabody Museum of Natural History (YPM) in New Haven, Connecticut. They were collected from the Beecher's Trilobite Bed site (Walcott Quarry: "Original Trilobite Bed" and beds 822 and 906) and Jo's Quarry in the Frankfort Shale of central New York State, and from Martin Quarry, at a slightly younger stratigraphic level, in the Whetstone Gulf Formation approximately 60 km to the north (localities detailed in [6, 7]). The stratigraphic units lie within the Maysvillian Regional Stage (Katian Stage), ~450 Ma before present [16]. The ostracods are preserved on pieces of mudstone; most are internal and external molds. There are also crumpled and incomplete valves and carapaces that are finely to coarsely pyritized. Some specimens preserve pyritized limbs, embryos, and other soft tissues. The supposed ostracods from the Lorraine Group held by the Peabody Museum total ~110 specimens: YPM 223936, 227953, 237106, 237107, 307286–307303, 505528, 505531–505537, 506492, 506506, 506521, 506532, 516164, 516170–516172, 516174–516196, 516197, 516199, 516200, 516201, 516203–516209, 516211–516233, 611120–611124, 611127, 611129, 611140–611142.

Etymology

Luprisca incuba is a new genus and species assigned to the Myodocopa [17] and provisionally to the Myodocopida. Its name alludes to the occurrence of brood care, as in the holotype (Lucina [goddess of childbirth] plus priscus [ancient] and incubatio [sitting on eggs]).

Material

Four open carapaces with soft parts were used: YPM 307293 (bed 906; Figures 1B, 1F, and 1G), YPM 307300 (holotype, Original Trilobite Bed; Figures 1A and 1D), and YPM 611120 (Original Trilobite Bed; Figure 1E) from Walcott Quarry and YPM 237107 (Figures 1C and 2A–2F; Movies S1 and S2 available online) from Martin Quarry. Each carapace is deformed by flattening and partially covered by matrix. Soft

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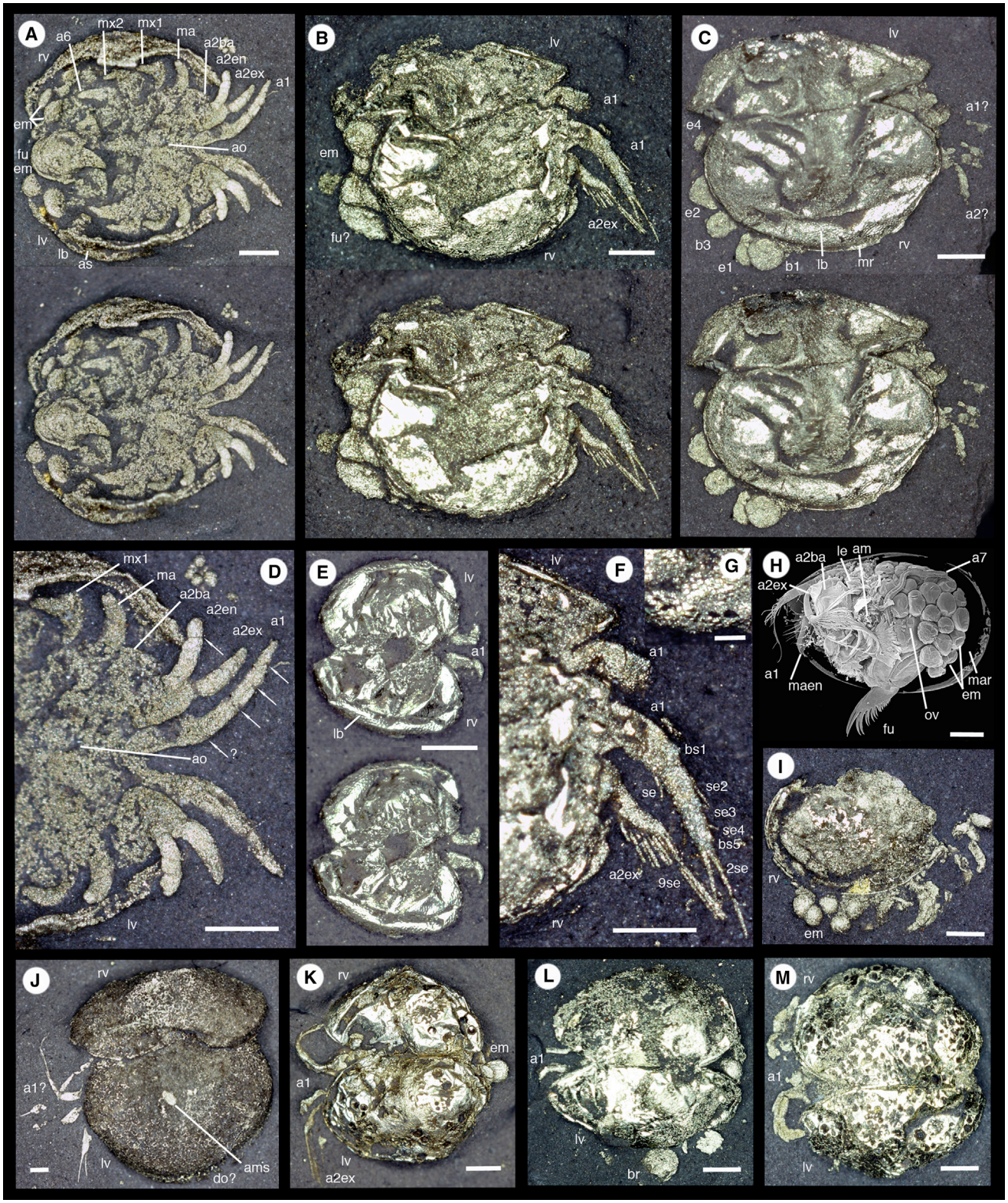


Figure 1. Ordovician and Extant Ostracods with Soft Parts

(A–G) *Luprisca incuba* gen. et sp. nov.

(A and D) Holotype, an open carapace with brood (YPM 307300; Original Trilobite Bed, Walcott Quarry), in ventral views: (A) a complete specimen (stereo pair); (D) detail of the anterior appendages.

(B, F, and G) Distorted open carapace (YPM 307293; bed 906, Walcott Quarry), in external right lateral views: (B) a complete specimen (stereo pair); (F) detail of the anterior appendages; (G) detail of the shell reticulation.

(C) Distorted open carapace with brood (YPM 237107; Martin Quarry), in external right lateral oblique view (stereo pair).

(legend continued on next page)

parts are best exposed in the holotype, but fine details are mostly masked by pyrite.

Diagnosis

A myodocope with a finely reticulate shell has a straight dorsal margin and a well-developed lateroadmarginal bend. The greatest height of the shell lies in front of the midlength. First and second appendages are clearly differentiated. Appendages 3–6 are elongate, gradually tapering and curve gently in a posterolateral direction.

Description

The carapace is highest just anterior of midlength; pervasive crumpling suggests that it was thin and/or poorly calcified. The maximum valve lengths (estimated or extrapolated where necessary) of the four known specimens are 2.5 mm (holotype), 2.4 mm (YPM 237107), 2.1 mm (YPM 307293), and 1.2 mm (YPM 611120). The valve dorsal margin is straight and the outline gently rounded posteriorly, ventrally, and anteriorly, and the outline of one specimen is weakly concave anteroventrally (YPM 237107; Figure 1C), although this is likely taphonomic. There is a well-developed lateroadmarginal bend, with a steeply inclined admarginal surface and a fine ridge along the free margin (Figures 1A–1C and 1E). A probable furrow runs immediately above and subparallel to the lateroadmarginal bend (YPM 237107 and juvenile YPM 611120; Figures 1C and 1D), and a possible adductorial sulcus is evident in both valves of YPM 237107 (Figures 1C and 2E); however, such features may be a result of crumpling of the carapace. The external surface of the valves of adults and the juvenile is finely reticulate to reticulostriate (Figures 1B, 1C, and 1E–1G).

The first antenna ("antennule") is stout, tapers gradually, and is comprised of at least five podomeres (Figures 1A, 1B, 1D, and 1F); podomere boundaries (evident in particular lighting conditions) are weakly defined. The podomeres decrease in size distally and have a combined length of ~1.2 mm in the holotype and ~0.9 mm beyond the carapace, excluding distal setae, in YPM 307293. The outer margin of each podomere bears a tiny slender seta (evident in YPM 307293 and preserved as basal stubs in three podomeres in the holotype), and a tiny seta is also present on the inner margin of possibly the fourth podomere from the end of the ramus (YPM 307293; Figures 1B and 1F). The first antenna ends in two prominent slender setae ~0.5 mm long (evident only in YPM 307293). As exposed in the holotype, the second antenna ("antenna") consists of a large subovoid basipod (maximum length 0.45 mm) and an exopod and slightly shorter endopod (0.92 mm and 0.73 mm long, respectively) (Figures 1A and 1D). Both rami taper gradually and distally, and, together with the first antenna, they curve gently and anterolaterally beyond the carapace. Nine closely set, slender spines are present on the exposed part of the exopod (evident only in YPM 307293; Figures 1B and 1F), including two distally. This setal pattern may reflect the arrangement of podomeres; setae provide evidence of articulations that are obscured by pyritization (there is typically one seta on podomeres 2–8 in extant cylindroleberidid myodocopes, with several on the terminal ninth podomere). Several faint, closely set transverse lines are evident in the distal part of the right exopod of the holotype (Figure 1D), which may mark podomere boundaries. The endopod reveals what are probably the terminal and penultimate podomeres (in the holotype only; Figure 1D).

The third, fourth, fifth, and sixth appendages (presumed mandible, first and second maxillae, and sixth limb; Figures 1A and 1D) in the holotype are mostly masked by pyritization; therefore, details of their morphology and possible differentiation are uncertain. Each appendage is elongate, tapers gradually and curves gently in a posterolateral direction (third appendage) to more posteriorly (fourth to sixth appendages) within the carapace; individual rami cannot be discerned. Adjacent and adaxial to the base of the presumed mandible and first maxilla, there is a weak elongate swelling with a small central cavity, the presumed atrium oris (Figures 1A and 1D). A slightly curved, hollow vermiform structure, with finely crenulated margins, may represent the seventh appendage (evident only in YPM 237107; Figures 2B–2D and 2F; Movies S1 and S2). The furca is well developed, consisting of two divergent lamelliform rami, but claws are not evident (Figure 1A). The curved triangular structure abaxially adjacent to the left claw may represent the displaced distal part of the right claw. A narrow elevation, presumably the body region, extends from the furca to immediately behind the atrium oris (Figure 1A).

The posterior part of the domicilium contains round-to-elongate ovoid and bivalved structures interpreted as brood. At least four partly exposed embryos are present in the holotype, one of which is ~0.24 mm in diameter (Figure 1A). Six embryos (one ~0.25 mm long) and four probable bivalved individuals (one ~0.35 mm long) interpreted as late-stage embryos or hatched juveniles are present in YPM 237107 (Figures 1C and 2A–2F; Movies S1 and S2; individual "b1" has a wide carapace gape and has probably hatched). One of the nonbivalved embryos occurs in an elongate posteroventral structure that is presumed to be an ovary (Figures 2C and 2D), and thus, it represents a very early developmental stage. Approximately four embryos are evident in YPM 307293 (Figure 1B).

Other Species

At least two other ostracod species, probably new, are represented by incompletely preserved specimens from the Lorraine Group.

(E) Small, distorted open carapace (YPM 611120; Original Trilobite Bed, Walcott Quarry), with anterior to the right (stereo pair).
(H) *Vargula hilgendorfii* with brood, a left lateral view with the left valve removed; extant from a baited trap, (July 2010), Asamushi, Aomori Prefecture, Japan.
(I) Ostracod sp. indet., an incomplete and distorted carapace with brood (YPM 516164; Original Trilobite Bed, Walcott Quarry), in right lateral view.
(J) Ostracod sp. indet. 1, a distorted open carapace mold (YPM 227953; Martin Quarry), in external left lateral oblique view.
(K) Ostracod sp. indet. 2, a distorted open carapace (YPM 307287; bed 906, Walcott Quarry), in external oblique dorsal view.
(L) Ostracod sp. indet. 2?, a distorted open carapace (YPM 611129; Original Trilobite Bed, Walcott Quarry), in external oblique dorsal view.
(M) Ostracod sp. indet., a distorted open carapace (YPM 237106; Martin Quarry), in external oblique dorsal view.
The following abbreviations were used: a1, first antenna; a2, second antenna; a2ba, basipod of second antenna; a2en, endopod of second antenna; a2ex, exopod of second antenna; a6, sixth limb; am, adductor muscle; ams, adductor muscle scar; ao, atrium oris; as, admarginal surface; b1 and b3, bivalved individuals; br, brachiopod; bs1 and bs5, base of seta; do?, dolon?; e1, e2, and e4, embryos; em, embryo(s); fu, furca; le, lateral eye; lb, lateroadmarginal bend; lv, left valve; ma, mandible; maen, endopod of mandible; mar, marsupium; mr, marginal ridge; mx1, first maxilla; mx2, second maxilla; rv, right valve; se, seta; se2–se4, setae; 2se and 9se, number of setae. Arrows mark presumed podomere boundaries.
Scale bars of (A)–(F) and (H)–(M) represent 500 μm; scale bar of (G) represents 75 μm. See also Movie S1.

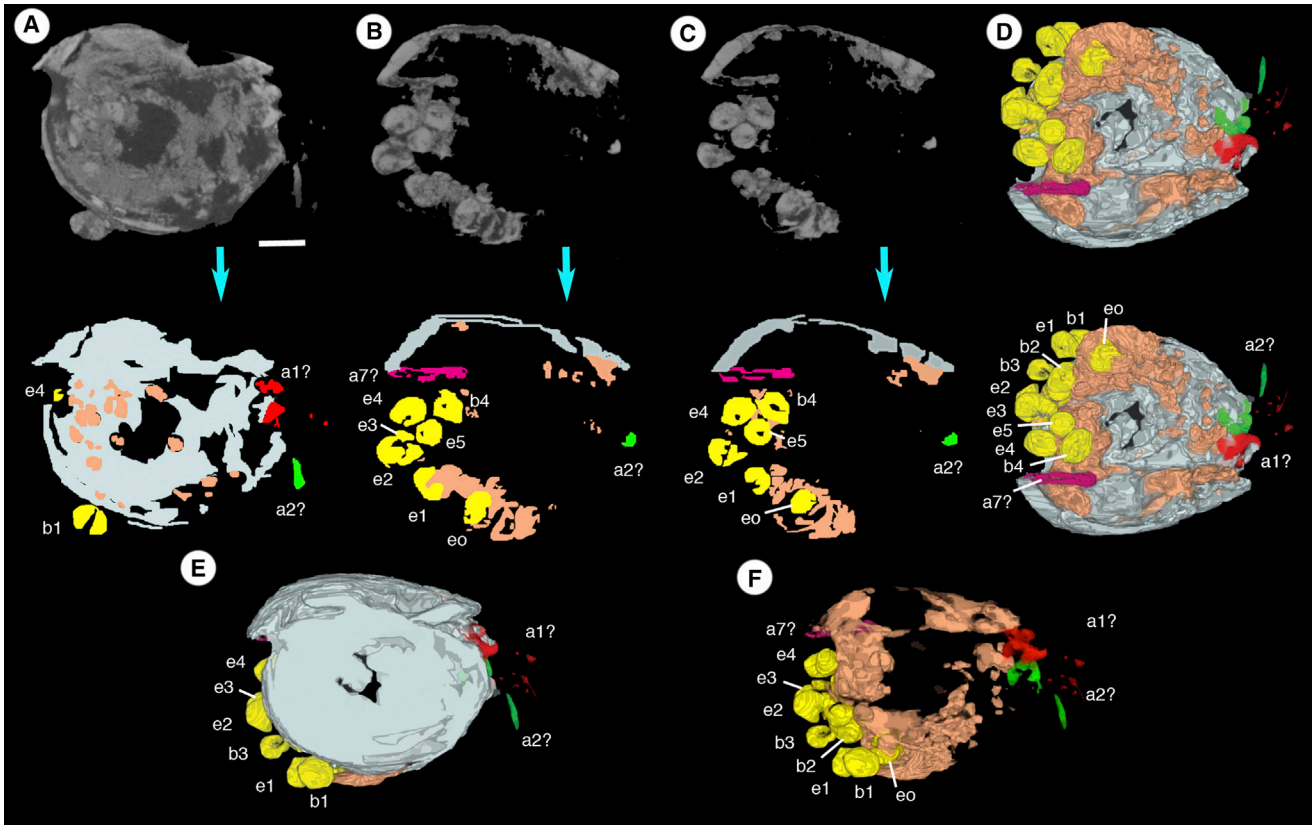


Figure 2. *Luprisca incuba* Gen. Et Sp. Nov., Distorted Open Carapace YPM 237107; from Martin Quarry

(A–C) Raw images (above) and interpretation (below) of CT slices 0093, 0119, and 0123 (Movies S1 and S2).

(D–F) Volume-rendered images: (D) an oblique ventral view (stereo pair); (E) an oblique right lateral view with the carapace; (F) an oblique right lateral view with the carapace omitted. The area that is gray is mostly shell; the body area is brown.

The following abbreviations were used: a7, seventh limb; e1–e5, embryos; b1–b4, bivalved individuals; eo, embryo in ovary. All other abbreviations are described in Figure 1 legend.

Scale bars of (A)–(F) represent 500 μm. See also Movie S2.

Ostracod sp. indet. 1 (YPM 227953, Martin Quarry; Figure 1J): The valve is very large and subamplete, with a central, subovate muscle scar. The presumed first antenna comprises a long, well-developed proximal podomere and six to seven much shorter podomeres; it bears several long, fine setae distally. The left valve preserves a long, narrow, adventral dolon-like structure.

Ostracod sp. indet. 2 (YPM 307287, bed 906, Walcott Quarry; Figure 1K): The valve is gently domed overall, with a shallow sulcus at midlength; it is more tumid posteriorly. The presumed first antenna is stout, and the presumed second antenna exopod comprises long proximal podomeres with finer podomeres and setae distally. Specimens YPM 307298 (bed 906, Walcott Quarry; with embryos), YPM 516199 (bed 906, Walcott Quarry), and YPM 611129 (Original Trilobite Bed, Walcott Quarry; Figure 1L) may also represent this species.

Ostracods spp. indet: Specimens YPM 223936 (Jo's Quarry; see Figure 2C in [7]) and YPM 237106 (Martin Quarry; Figure 1M) preserve just the first antenna, which, like that of *L. incuba*, is strongly developed. The first antenna of YPM 237106 consists of at least five podomeres, as in *L. incuba*, each with a setal base. Specimen YPM 516164 (Original Trilobite Bed, Walcott Quarry; Figure 1I) preserves unresolved frontal and other appendages and a brood of at least five embryos beyond the carapace.

Many of the valves, especially of the indeterminate species, show polygonal-to-circular and radiate microstructures (e.g., Figures 1K–1M) identical to examples described from Silurian myodocopids [18]. As compared to mineralization processes in living myodocopes, the Silurian examples were interpreted to be the surface expression of calcium carbonate platelets. They are probably the result of in vivo shell calcification and postmortem diagenetic changes, including recrystallization, which produced acicular radii in platelets of various sizes.

Discussion

Approximately nine ostracod species were reported from the Lorraine Group in the early 19th century [19]. They include palaeocopes, binodicopes, and leiocopes, but none bear close similarity to *L. incuba.* Some of the indeterminate species documented herein have aparchitid-like valves (e.g., Figures 1K–1M). In 1973, a low abundance of *Aparchites* sp. and *Primitia* sp. was noted among supposed ostracods from the Beecher's Trilobite Bed [20], and, more recently, preserved appendages were reported from the locality [6]—the first ever record of ostracods with soft parts preserved in pyrite.

The New Myodocope Genus

The morphology preserved in *L. incuba* confirms the presence of ostracods in the Ordovician and extends our knowledge of

the paleobiology of the group back some 25 million years from the mid-Silurian Herefordshire Lagerstätte [1–4]. *L. incuba* is assigned here to the Myodocopa and provisionally to the Myodocopida. The podomere and setal arrangement of the first antenna is the same as that of extant Myodocopida, even down to the tiny ventral seta and two long distal setae, which are comparable to the sensory bristle on the fifth podomere and the "C" and "F" bristles of the seventh and eighth (distalmost) podomeres, respectively, of many cylindroleberidid myodocopids (see [21]). A similar pair of distal setae is also present in the first antenna of the Silurian myodocopids *Colymbosathon*, *Nymphatelina*, and *Nasunaris* [1–3]. The second antenna of *L. incuba*, with its large ovoid basipod and an exopod longer than the endopod, also accords with the morphology of myodocopes rather than podocopes (in which the endopod is better developed). Moreover, if the position of the setae on the exopod of the second antenna reflects the arrangement of podomeres, the total number mirrors that in extant myodocopes (the exopod in the majority of myodocopids and halocyprids consists of nine podomeres and rarely less) compared to one or two podomeres, or just setae, in podocopes [22]. The slender, vermiform morphology of the putative seventh appendage in *L. incuba* is known only in myodocopids. The strongly sclerotized nature of the furca also conforms more to that of extant myodocopes than podocopes, as does the probable thin and/or poorly calcified nature of the shell. In view of these similarities, it is striking that the external morphology of the shell shows no defining myodocope feature, such as a rostrum or rostral incisure. It is clear that caution is required when classifying ostracods based on the carapace alone [4]. The one other known supposed Ordovician myodocope, recorded only from shells [23], is *Myodoprimigenia fistuca* from Hirnantian-age rocks (~443 Ma) of South Africa.

Brooding Strategy

The brooding strategy evident in *L. incuba* accords with that of extant myodocopids (Figure 1H). It allows the gender of several specimens to be determined in this Ordovician species and provides fossil evidence that a brooding reproductive mode has persisted in myodocopes for at least 450 Ma. This reproductive strategy may be even older, given that molecular phylogenetics predicts that myodocopes occur in the Cambrian period [14]. Eggs of fossil invertebrates, especially putative in situ embryos, are very rare, including possible examples of eggs and brooding in fossil ostracods [2]. Most supposed fossil invertebrate eggs are recovered isolated from the assumed parent. Putative in situ eggs or embryos have been found in fossils of a bradoriid arthropod species [24], a tealliocaridid malacostracan, a syncarid malacostracan, two branchiopod species, and a few ostracod species ([2] and references therein).

Most extant ostracods are Podocopa, the majority of which reproduce sexually and deposit their eggs on a substrate, whereas most Myodocopa (all myodocopids and at least one halocyprid), Platycopa, and Darwinuloidea brood embryos in the posterior area of the domicilium [25, 26]. The brood number and size of the embryos in *L. incuba* is similar to that in extant, sexually mature, brooding myodocopes [25]. The concurrence of ovarian and brooded embryos (i.e., clutch overlap) is also known in some extant myodocopes [25]. *L. incuba* is unique in providing evidence of ovarian, embryonic, and possible juvenile developmental stages in a fossil invertebrate. In the extant myodocopid *Vargula hilgendorfii*, embryonic development takes about 16 days, from egg on the ovary wall [27] to hatched limb-bearing bivalved juvenile [28, 29]. The only other unequivocal example of eggs in a fossil ostracod is in a specimen of the mid-Silurian myodocope *Nymphatelina gravida* from the Herefordshire Konservat-Lagerstätte (~425 Ma), whose in situ brood includes a bivalved individual [2]. Evidence of reproductive strategies in fossil ostracods is otherwise indirect: presumed sexual dimorphism of the shell in many groups involves what is possibly brood space or, rarely, the supposed occurrence of only presumed females, suggesting parthenogenesis ([2] and references therein).

Habitat and Lifestyle

The Lorraine Group accumulated in the relatively deep axis of a foreland basin that formed on the margin of the paleocontinent Laurentia during the Cambrian-Ordovician Taconic orogeny. The ostracods occur in thin gray mudstones interpreted as low-density distal turbidites that entrained and rapidly buried a low-diversity/low-abundance benthic fauna that includes *Triarthrus* with preserved appendages. The assemblage is interpreted as living in dysoxic bottom-water conditions [6, 7, 30]. The association of ostracods and *Triarthrus* here and elsewhere in the Taconic foreland basin [19, 31, 32] shows that ostracods occur in deep-water associations with olenid trilobites during the Ordovician period (contra [33]). Conjoined ostracod valves are common, indicating that they were probably not subject to prolonged high-energy conditions postmortem. *L. incuba* has well-developed frontal appendages, suggesting that it was a swimmer. Most extant myodocopes are nektobenthic predators, scavengers, or detritivores [34], and the lifestyle of *L. incuba* was probably similar.

Experimental Procedures

The fossils were prepared using mechanical techniques, including fine air abrasion. All except the extant ostracod (Figure 1H; a scanning electron micrograph) were photographed under alcohol using a Nikon Multiphot stand and associated bellows equipment mounted with a Canon EOS 5D camera (Figures 1A–1G and 1I–1M), following the methods of [35]. Images of specimen YPM 237107 were also captured by computed tomography (CT) (Figures 2A–2C; Movies S1 and S2), and volume was rendered and laterally emphasized (400×) in order to compensate for flattening of the fossil due to compaction (Figures 2D–2F). The figured CT images were obtained using a ScanXmate-L system (Comscantecno) at a resolution of 20.51 μm at 151 kV. Some 260 tiff images were captured, each 683 × 604 pixels in size with a resolution of 0.51 pixel/cm. Using contrasts in density, soft parts (thick density) and shell (thin density) were distinguished and colored (RGB). ImageJ software [36] was used to convert the 2D CT image stacks into a 3D volumetric image. Volume rendering was performed using virtual stack intervals at 20.51 μm, with threshold 0 and resampling factor 2.

Supplemental Information

Supplemental Information includes two movies and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.02.040.

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